

Ethan Chang

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Professional Summary

Software Engineer with experience in machine learning, data engineering, and cloud computing driven by personal growth. Passionate about solving challenging problems alongside collaborative and experienced professionals. Comfortable working in fast-paced environments and producing tangible results such as building and deploying scalable data pipelines in Snowflake, designing and training machine learning models using Pytorch, and architecting cloud-based infrastructure in Azure. Currently completing an M.Sc. in Computer Science at UIUC to deepen expertise in AI and cybersecurity.

Education

University of Illinois at Urbana-Champaign, Champaign, IL

Jan. 2025 – Dec. 2026

M.Sc. Computer Science | Honors: 3.9/4.0 GPA

Coursework: Cloud Computing Apps, Machine Learning, Computer Security 1 & 2, DL For Healthcare

Northeastern University, Boston, MA

Sep. 2020 – May. 2024

B.Sc. Computer Science & Business Administration | Honors: 3.7/4.0 GPA, Magna Cum Laude, Dean's List 2020-2024

Coursework: Foundations of AI, Machine Learning in Finance, Networks & Distributed Sys, Database Management Sys

Professional Experience

Software Engineer, Full-time Co-op, Motorola Solutions – Boston, MA

Jan. 2023 - Jun. 2023

- Reduced the magnitude of traffic spikes by 33% for the core video API service saving on cloud costs and reducing latency
- Enabled seamless transition to Avigilon's new cloud platform for legacy devices by migrating customer account data
- Streamlined the CI/CD process by adding additional integration tests and sending release requests for QA and Production environments with Azure Pipelines

Data Engineer, Full-time Co-op, The Travelers Company – Hartford, CT

Jan. 2022 - Aug. 2022

- Spearheaded the cloud migration initiative from on-prem data servers to the Snowflake platform for cost reduction and to expedite OLAP queries for underwriting
- Built an ETL pipeline to clean over 70 million rows of policy data and migrate it to Snowflake
- Provided guidance for colleagues about the Snowflake platform, how to use the ETL pipeline, and locate documentation to continue the cloud migration

Technical Projects

Bank Churn Predictive Modelling, Python, Keras, NLTK, Pandas, Scikit-Learn, Seaborn, Matplotlib

Apr. 2024

- Led a team of four to evaluate whether banking institutions could predict customer churn to facilitate proactive customer retention strategies
- Identified key requirements, critically a low false negative rate, and trained models accordingly achieving an F1-Scores of 0.91
- Presented the results to a panel of C-suite executives with expertise in entrepreneurship and financial technology

Faux Formicidae Genetic Model, Python, PyTorch, OpenCV, Pygame

Jan. 2024 - Apr. 2024

- Simulated a virtual ant colony replicating real-world foraging behavior and pheromone communication using a modified Ant Colony Optimization algorithm
- Compared multiple models each in charge of a colony by assessing the differing emergent behaviors and optimal end states

Technical Skills

Languages: Python, C#, Java, SQL, R, Bash, PowerShell, Mongosh, Cypher, Dlang, LaTeX, Racket (Lisp)

Technologies: AWS, MS Azure, Kubernetes, Docker, Apache Spark, GitHub, Git, Jira, Confluence

Development Tools: Visual Studio Code, SQL SMS, Tableau, VirtualBox, RStudio, IBM SPSS, Talend

OS: Windows, Ubuntu Linux, MacOS